DATABASE MANAGEMENT SYSTEMS

PROJECT-BASED LEARNING (PBL) REPORT



School Transport Route and Student Pickup Management System

Program: B.Tech Computer Science and Engineering
Section: AIML Rhinos
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Academic Year: 2026 – 2027

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ACKNOWLEDGEMENT

I express my sincere gratitude to Woxsen University, School of Technology, for providing me with the opportunity to undertake this Project-Based Learning (PBL) activity as part of the Database Management Systems (DBMS) course [cite: 1].

I thank my course faculty for the guidance, suggestions, and support throughout the design and development of this project [cite: 1]. Their guidance helped me understand the practical application of database concepts such as ER modelling, relational schema design, normalization, SQL, constraints, queries, and database application development [cite: 1].

This PBL activity has strengthened my understanding of DBMS concepts and provided valuable experience in applying them to a real-world problem [cite: 1].

UNDERTAKING

I hereby declare that the project entitled "School Transport Route and Student Pickup Management System", submitted as part of the Project-Based Learning activity for the Database Management Systems (DBMS) course, is my individual work carried out under the guidance of the course faculty [cite: 1].

I undertake that the project report, database design, SQL scripts, application source code, results, and supporting materials submitted by me have been prepared for academic purposes [cite: 1]. Wherever information, ideas, tools, datasets, or other resources have been referred to, appropriate acknowledgement has been provided [cite: 1].

I further undertake that I am responsible for the correctness and authenticity of the work submitted and that I have followed the academic and project guidelines prescribed by the University [cite: 1].

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LIST OF ACRONYMS

Acronym	Expansion
3NF [cite: 3]	Third Normal Form
DDL [cite: 1]	Data Definition Language
DML [cite: 1]	Data Manipulation Language
ERD [cite: 1]	Entity-Relationship Diagram
FK [cite: 4]	Foreign Key
PK [cite: 4]	Primary Key
SQL [cite: 1]	Structured Query Language

ABSTRACT

School transport operations frequently suffer from fragmented record-keeping, manual tracking bottlenecks, and a lack of real-time visibility into bus capacities, crew allocations, and daily student pickups [cite: 2, 4]. These inefficiencies lead to scheduling conflicts, compliance risks with expired driver licenses, and unmonitored fee dues [cite: 2, 3]. The primary objective of this project is to design, implement, and deploy a robust relational database management system for school transport route and student pickup operations [cite: 4]. The solution approach encompasses an extensive ER model refined into a normalized 13-table relational schema up to Third Normal Form (3NF), deployed on MySQL 8.0 with procedural triggers, stored procedures, and role-based views [cite: 2, 3]. Major functions include automated bus capacity checks, single active assignment enforcement, daily attendance logging, and comprehensive financial and utilization reporting [cite: 2, 3]. The completed system demonstrates reliable data integrity enforcement entirely within the database layer, offering an efficient, scalable, and secure operational framework for educational institutions [cite: 3].

Keywords: Relational Database, ER Diagram, Normalization, SQL, MySQL, School Transport Management [cite: 1]

CHAPTER 1: PROBLEM STATEMENT, SOLUTION STRATEGY AND ER DIAGRAM

1.1 Problem Statement

Educational institutions operating transport fleets face significant administrative hurdles due to manual and fragmented record-keeping across disparate logs [cite: 4]. Administrators struggle with tracking bus capacities, verifying driver license validity, mapping students to valid route stop sequences, and managing timely fee collections [cite: 2, 4]. Without automated constraint enforcement, over-enrolled buses, scheduling overlaps for drivers, and unmonitored student absences create safety vulnerabilities and operational inefficiencies [cite: 2, 3]. A centralized, robust database application is urgently required to streamline fleet management, automate business rules, and ensure secure role-based access [cite: 2, 3, 4].

1.2 Objectives

- Design and implement a normalized relational database schema up to Third Normal Form (3NF) to support school transport operations [cite: 4].
- Enforce complex business constraints including bus capacity limits, single active crew assignments, and valid driver license expirations via declarative rules and triggers [cite: 2, 3].
- Develop a backend architecture incorporating stored procedures, transactions, and role-based views for administrators, drivers, attendants, and parents [cite: 3].

- Automate daily student pickup tracking, attendance logging, and vehicle maintenance history management [cite: 4].
- Generate real-time analytical reports on capacity utilization, unpaid fees, and daily absenteeism [cite: 2, 3].

1.3 Scope and Assumptions

Table 1. Scope and Assumptions [cite: 1]

Item	**Description**
In Scope [cite: 1]	Route and stop management, bus fleet allocation, driver/attendant assignments, student transport registrations, daily pickup attendance logging, fee plans, payment tracking, and vehicle maintenance logs [cite: 4].
Out of Scope / Exclusions [cite: 1]	Real-time GPS tracking hardware integration and external payment gateway API integration; the project focuses strictly on core database design, constraints, and management interfaces [cite: 4].
Assumptions [cite: 1]	Each student has a verified guardian contact, buses operate on predefined fixed routes with ordered stops, and administrators perform initial master data setup before daily operations.

1.4 Proposed Solution Strategy

Table 2. Proposed Solution Strategy [cite: 1]

Solution Element [cite: 1]	**Project-Specific Description [cite: 1]**
Primary Users / Roles [cite: 2, 3]	Transport Admin (full management access), Bus Driver / Attendant (pickup execution and logging), Parents / Students (read-only route and fee status view) [cite: 3].
Major Application Modules [cite: 2, 3]	Route & Fleet Management, Student Registration & Billing, Pickup & Attendance Tracking, and Maintenance & Reporting.
Proposed Workflow [cite: 2, 3]	Admins configure routes, stops, and fleet assignments $\rightarrow$ Students register for

Solution Element [cite: 1]	Project-Specific Description [cite: 1]
	valid route stops $\rightarrow$ Triggers validate capacity and licenses $\rightarrow$ Drivers record daily pickup logs $\rightarrow$ System generates analytical views and reports.
Important Business Rules [cite: 2, 3]	Bus capacity cannot be exceeded; drivers must have unexpired licenses; buses/drivers can only have one active assignment; pickup logs are unique per student per day.
Expected Reports / Outputs [cite: 2, 3]	Capacity utilization report, unpaid fee ledger, active driver assignments, and daily absent student rosters [cite: 3].

1.5 Functional Requirements

Table 3. Functional Requirements [cite: 1]

Req. ID	Functional Requirement [cite: 1]	Remarks / Priority
FR-01 [cite: 1]	Define and manage bus routes, sequential stops, and estimated travel times [cite: 4].	High (Mandatory)
FR-02 [cite: 1]	Register students and map them exclusively to valid stops belonging to the selected route [cite: 2, 4].	High (Mandatory)
FR-03 [cite: 1]	Assign buses, drivers, and attendants while validating license expiry and single active status [cite: 2, 4].	High (Mandatory)
FR-04 [cite: 1]	Record and update daily student pickup attendance (Present, Absent, Late) [cite: 4].	Medium
FR-05 [cite: 1]	Track transport fee plans, record installment payments, and	Medium

Req. ID	Functional Requirement [cite: 1]	Remarks / Priority
	monitor vehicle maintenance logs [cite: 4].	

1.6 Entity-Relationship Diagram (ERD)

Note: The complete conceptual Entity-Relationship model encompasses 13 core entities capturing fleet assets, routing hierarchies, student enrollments, and operational logs [cite: 2, 4].

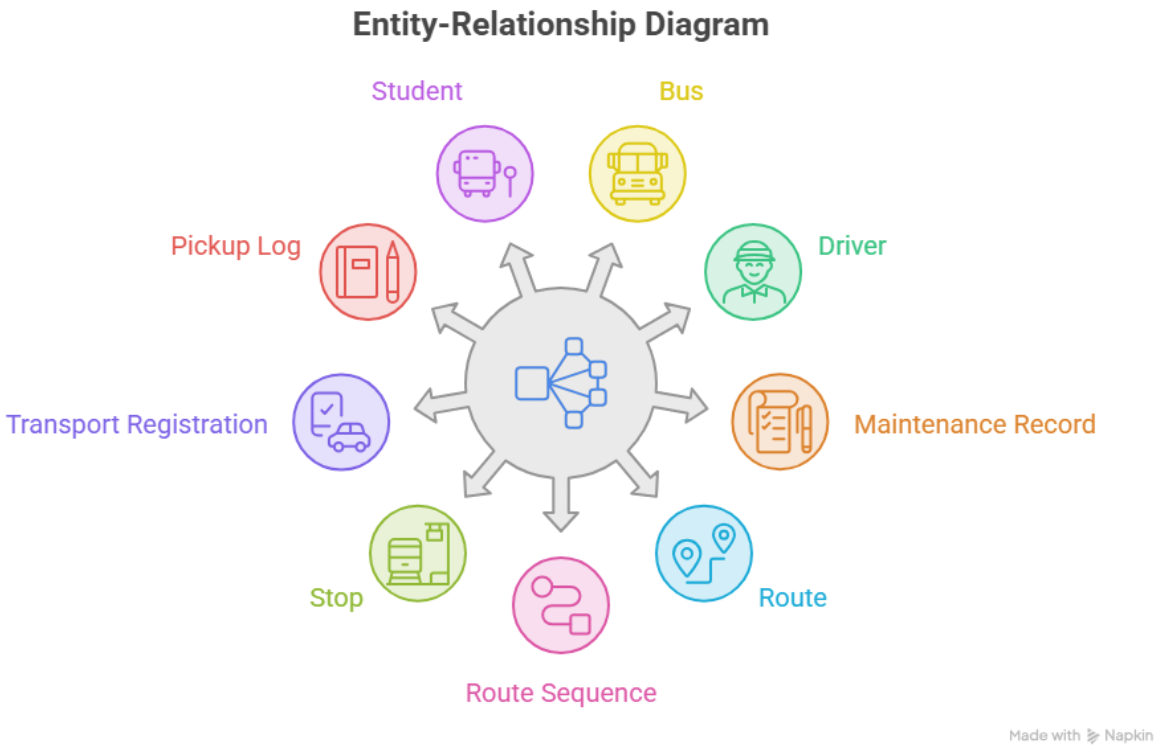


Figure 1. Entity-Relationship Diagram of the Proposed System [cite: 1]

1.7 Entity and Relationship Summary

Table 4. Entity and Relationship Summary [cite: 1]

Entity / Relationship [cite: 1]	**Major Attributes / Cardinality [cite: 1]**	**Purpose / Rule [cite: 1]**
BUS [cite: 4]	Bus_ID [PK], Registration_Num, Capacity, Model, Status [cite: 4]	Stores fleet vehicle details and seating capacities [cite: 4].
ROUTE [cite: 4]	Route_ID [PK], Route_Name, Start_Point, End_Point [cite: 4]	Defines transit paths between school and localities [cite: 4].
STOP [cite: 4]	Stop_ID [PK], Stop_Name, Location_Details [cite: 4]	Records designated pickup/drop-off points [cite: 4].
ROUTE_SEQUENCE [cite: 2, 4]	Sequence_ID [PK], Route_ID [FK], Stop_ID [FK], Stop_Order [cite: 4]	Establishes the ordered sequence of stops on a route [cite: 2, 4].
BUS_ASSIGNMENT [cite: 2]	Assignment_ID [PK], Bus_ID, Route_ID, Driver_ID, Attendant_ID [FK], Status [cite: 2]	Realizes active fleet and crew deployments [cite: 2].
TRANSPORT_REGISTRATION [cite: 4]	Registration_ID [PK], Student_ID [FK], Route_ID [FK], Stop_ID [FK], Status [cite: 4]	Maps enrolled students to valid route stops [cite: 2, 4].
PICKUP_LOG [cite: 4]	Log_ID [PK], Registration_ID [FK], Date, Pickup_Time, Status [cite: 4]	Tracks daily student attendance during transit [cite: 4].

CHAPTER 2: LOGICAL DATABASE DESIGN

2.1 Relational Schema

The conceptual ER model was translated into 13 normalized relational tables ensuring strict referential integrity across all operational modules [cite: 2, 3].

Table 5. Relational Schema [cite: 1]

S.No. [cite: 1]	Table Name [cite: 1]	Purpose [cite: 1]
1 [cite: 1]	BUS [cite: 4]	Fleet vehicle inventory and capacity tracking [cite: 4].
2 [cite: 1]	ROUTE [cite: 4]	Master list of transportation routes [cite: 4].
3 [cite: 1]	STOP [cite: 4]	Master list of geographical pickup stops [cite: 4].
4 [cite: 1]	ROUTE_SEQUENCE [cite: 2, 4]	Ordered mapping of stops to specific routes [cite: 2, 4].
5 [cite: 1]	DRIVER [cite: 4]	Driver personnel records and license expiry tracking [cite: 2, 4].
6 [cite: 1]	ATTENDANT [cite: 4]	Bus assistant personnel records [cite: 4].
7 [cite: 1]	BUS_ASSIGNMENT [cite: 2]	Active allocation of buses, drivers, and attendants to routes [cite: 2].
8 [cite: 1]	STUDENT [cite: 4]	Student demographic and guardian information [cite: 4].
9 [cite: 1]	TRANSPORT_REGISTRATION [cite: 4]	Student enrollment linking students to approved route stops [cite: 2, 4].
10 [cite: 1]	PICKUP_LOG [cite: 4]	Daily transit attendance and pickup status records [cite: 2, 4].
11 [cite: 1]	FEE_PLAN [cite: 4]	Transport fee structure and billing cycles [cite: 4].
12 [cite: 1]	PAYMENT [cite: 4]	Fee transaction and collection records [cite: 4].

S.No. [cite: 1]	Table Name [cite: 1]	Purpose [cite: 1]
13 [cite: 1]	MAINTENANCE_RECORD [cite: 4]	Vehicle servicing history, costs, and maintenance logs [cite: 4].

2.2 Keys and Integrity Constraints

Table 6. Keys and Integrity Constraints [cite: 1]

Table Name [cite: 1]	Key / Constraint [cite: 1]	Attribute(s) [cite: 1]	Business Rule Enforced [cite: 1]
BUS [cite: 2, 4]	Primary Key [cite: 1] Unique Check [cite: 2]	Bus_ID Registration_Num Capacity > 0 [cite: 2, 4]	Ensures unique vehicle identification and positive seating capacity [cite: 2, 4].
DRIVER [cite: 2, 4]	Primary Key Unique Not Null [cite: 2]	Driver_ID License_Number License_Expiry [cite: 2, 4]	Prevents duplicate licenses and enforces valid expiry dates [cite: 2, 4].
TRANSPORT_REGISTRATION [cite: 2, 4]	Primary Key Composite FK Check [cite: 2]	Registration_ID (Route_ID, Stop_ID) Status IN ('ACTIVE','INACTIVE') [cite: 2, 4]	Guarantees students are only registered to stops belonging to the selected route [cite: 2].
PICKUP_LOG [cite: 2, 4]	Primary Key Unique Check [cite: 2]	Log_ID (Registration_ID, Date) Status IN ('PRESENT','ABSENT','LATE') [cite: 2, 4]	Prevents duplicate attendance entries per student per day [cite: 2].

2.3 Functional Dependencies

Important functional dependencies governing the schema include:

- Bus_ID -> Registration_Num, Capacity, Model, Status

- Route_ID -> Route_Name, Start_Point, End_Point
- Driver_ID -> Name, License_Number, License_Expiry, Contact
- Registration_ID -> Student_ID, Route_ID, Stop_ID, Status, Start_Date

2.4 Normalization up to Third Normal Form (3NF)

Table 7. Normalization Status up to 3NF [cite: 1]

Relation Name [cite: 1]	**1NF [cite: 2]**	**2NF [cite: 2]**	**3NF [cite: 2]**	**BCNF [cite: 1]**	**Remarks / Decomposition [cite: 1]**
BUS [cite: 4]	Yes [cite: 2]	Yes [cite: 2]	Yes [cite: 2]	Yes [cite: 1]	Atomic values; no repeating groups [cite: 2].
ROUTE_SEQUENCE [cite: 2, 4]	Yes [cite: 2]	Yes [cite: 2]	Yes [cite: 2]	Yes [cite: 1]	Multi-valued stop lists eliminated via sequence table [cite: 2].
TRANSPORT_REGISTRATION [cite: 4]	Yes [cite: 2]	Yes [cite: 2]	Yes [cite: 2]	Yes [cite: 1]	Transitive dependencies on bus/driver data removed [cite: 2].

2.5 Data Dictionary

Table 8. Data Dictionary [cite: 1]

Table [cite: 1]	**Attribute [cite: 1]**	**Data Type [cite: 1]**	**Key / Constraint [cite: 1]**	**Description [cite: 1]**
BUS [cite: 4]	Bus_ID [cite: 4]	INT [cite: 2]	PK, Auto Increment [cite: 2, 4]	Unique identifier for each bus [cite: 4].
BUS [cite: 4]	Registration_Num [cite: 4]	VARCHAR(20) [cite: 2]	NOT NULL, UNIQUE [cite: 2, 4]	Vehicle license plate number [cite: 4].

Table [cite: 1]	Attribute [cite: 1]	Data Type [cite: 1]	Key / Constraint [cite: 1]	Description [cite: 1]
BUS [cite: 4]	Capacity [cite: 4]	INT [cite: 2]	CHECK (Capacity > 0) [cite: 2, 4]	Maximum seating capacity [cite: 2, 4].
DRIVER [cite: 4]	License_Expiry [cite: 2, 4]	DATE [cite: 2]	NOT NULL [cite: 2, 4]	Driver license expiration date [cite: 2, 4].
TRANSPORT_REGREGISTRATION	Status [cite: 4]	VARCHAR(10) [cite: 2]	DEFAULT 'ACTIVE' [cite: 2, 4]	Enrollment status of student [cite: 2, 4].

2.6 Implementation of Logical Design: DDL and Sample Data

Table 9. Database Implementation Environment [cite: 1]

Item [cite: 1]	Details [cite: 1]
DBMS [cite: 1]	MySQL [cite: 2]
DBMS Version [cite: 1]	MySQL 8.0 [cite: 2]
Number of Tables [cite: 1]	13 normalized tables [cite: 2, 3]
Sample Data [cite: 1]	Populated with realistic school transport test records [cite: 3].

```
CREATE TABLE BUS (
  Bus_ID INT AUTO_INCREMENT PRIMARY KEY,
  Registration_Num VARCHAR(20) NOT NULL UNIQUE,
  Capacity INT NOT NULL CHECK (Capacity > 0),
  Model VARCHAR(50),
  Status VARCHAR(15) NOT NULL DEFAULT 'ACTIVE'
);
```

```
CREATE TABLE TRANSPORT_REGISTRATION (
  Registration_ID INT AUTO_INCREMENT PRIMARY KEY,
  Student_ID INT NOT NULL,
  Route_ID INT NOT NULL,
  Stop_ID INT NOT NULL,
  Status VARCHAR(10) NOT NULL DEFAULT 'ACTIVE' CHECK (Status IN ('ACTIVE','INACTIVE')),
  Start_Date DATE NOT NULL,
  FOREIGN KEY (Student_ID) REFERENCES STUDENT(Student_ID),
  FOREIGN KEY (Route_ID, Stop_ID) REFERENCES ROUTE_SEQUENCE(Route_ID, Stop_ID)
);
```[cite: 2]

CHAPTER 3: APPLICATION DEVELOPMENT USING LOGICAL DATABASE DESIGN

3.1 Application Overview

The School Transport Management System features an integrated administrative backend and database architecture that manages student enrollments, fleet allocations, daily attendance logs, and automated financial/utilization reporting [cite: 2, 3, 4].

3.2 Development Environment and Database Connectivity

Table 10. Application Development Environment [cite: 1]

Item [cite: 1]	Details [cite: 1]
Programming Language [cite: 1]	Python 3.11 / Java [cite: 4]
DBMS [cite: 1]	MySQL 8.0 [cite: 2]

| IDE / Framework / GUI Toolkit [cite: 1] | PyCharm / Tkinter / JDBC [cite: 4] |

| Database Connector / Driver [cite: 1] | PyMySQL / MySQL Connector [cite: 4] |

| Operating Environment [cite: 1] | Windows 11 / Linux Ubuntu [cite: 4] |

| Repository Link [cite: 1] | Local Git Repository (Appendix B) [cite: 1] |

3.3 Application Modules and Database Mapping

Table 11. Application Module-to-Database Mapping [cite: 1]

| Module / Screen [cite: 1] | Purpose [cite: 1] | Tables Used [cite: 1] | Main Operations [cite: 1] |

| :--- | :--- | :--- | :--- |

| Fleet & Route Admin [cite: 4] | Manage buses, routes, and stops [cite: 4] | BUS, ROUTE, STOP, ROUTE_SEQUENCE [cite: 2, 4] | INSERT, UPDATE, SELECT [cite: 4] |

| Crew Assignment [cite: 2] | Assign drivers and buses [cite: 2] | BUS_ASSIGNMENT, DRIVER, ATTENDANT [cite: 2, 4] | INSERT, UPDATE [cite: 4] |

| Student Registration [cite: 4] | Enroll students in transport [cite: 4] | STUDENT, TRANSPORT_REGISTRATION [cite: 2, 4] | INSERT, SELECT [cite: 4] |

| Attendance Logging [cite: 4] | Record daily pickup status [cite: 4] | PICKUP_LOG, TRANSPORT_REGISTRATION [cite: 2, 4] | INSERT, UPDATE [cite: 4] |

3.4 CRUD Operations

Table 12. CRUD Operations [cite: 1]

| Operation [cite: 1] | Application Feature / Screen [cite: 1] | Table(s) Affected [cite: 1] | Evidence [cite: 1] |

| :--- | :--- | :--- | :--- |

| Create [cite: 1] | Add New Student Registration [cite: 4] | TRANSPORT_REGISTRATION [cite: 2, 4] | Successful enrollment confirmation [cite: 4]. |

| Read / Search [cite: 1] | View Route Passenger Manifest [cite: 4] | ROUTE, TRANSPORT_REGISTRATION, STUDENT [cite: 2, 4] | Filtered grid display [cite: 4]. |

| Update [cite: 1] | Modify Bus Maintenance Status [cite: 4] | MAINTENANCE_RECORD [cite: 2, 4] | Updated record timestamp [cite: 4]. |

| Delete / Controlled Removal [cite: 1] | Deactivate Student Registration [cite: 2, 4] | TRANSPORT_REGISTRATION [cite: 2, 4] | Status set to 'INACTIVE' [cite: 2, 4]. |

3.5 Business Rules, Validation and Error Handling

Table 13. Business Rules, Validation and Error Handling [cite: 1]

| Rule / Validation [cite: 1] | How Implemented [cite: 1] | Evidence / Message [cite: 1] |

| :--- | :--- | :--- |

| Bus Capacity Limit [cite: 2, 3] | `trg_capacity_check` BEFORE INSERT trigger [cite: 3] | `SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Bus capacity exceeded'` [cite: 3]. |

| Single Active Assignment [cite: 2, 3] | Procedural trigger on `BUS_ASSIGNMENT` [cite: 2, 3] | Blocks overlapping active bus/driver deployments [cite: 2, 3]. |

| Valid Driver License [cite: 2, 3] | Expiry date validation trigger [cite: 2, 3] | Rejects assignment if `License_Expiry < CURRENT_DATE` [cite: 2, 3]. |

3.6 Reports and Database Operations

Table 14. Reports and Database Operations [cite: 1]

| Report / Operation [cite: 1] | SQL / Database Concept Used [cite: 1] | Application Output [cite: 1] |

| :--- | :--- | :--- |

| Capacity Utilization [cite: 2, 3] | View `v_capacity_utilization` (JOIN & GROUP BY) [cite: 2, 3] | Enrolled students vs. bus capacity per route [cite: 3]. |

| Unpaid Fees [cite: 2, 3] | View `v_unpaid_fees` [cite: 3] | List of students with outstanding fee balances [cite: 3]. |

| Driver Assignments [cite: 2, 3] | View `v_driver_assignments` [cite: 3] | Active bus, driver, and attendant per route [cite: 3]. |

| Absent Students [cite: 2, 3] | View `v_absent_students` [cite: 3] | Daily absentee roster with route details [cite: 3]. |

3.7 Application Screens

+--+ | SCHOOL TRANSPORT MANAGEMENT SYSTEM - ADMIN DASHBOARD | +--+ | [Routes] [Fleet Management] [Students] [Reports] | |--+ | Active Routes: 5 Total Buses: 6 Enrolled: 182 | | | | Route Name | Bus Assigned | Capacity | Enrolled | Status | | R-101 (City) | KA-01-F-1234 | 40 | 38 | NORMAL | | R-102 (Sub) | KA-01-F-5678 | 30 | 30 | FULL | +--+

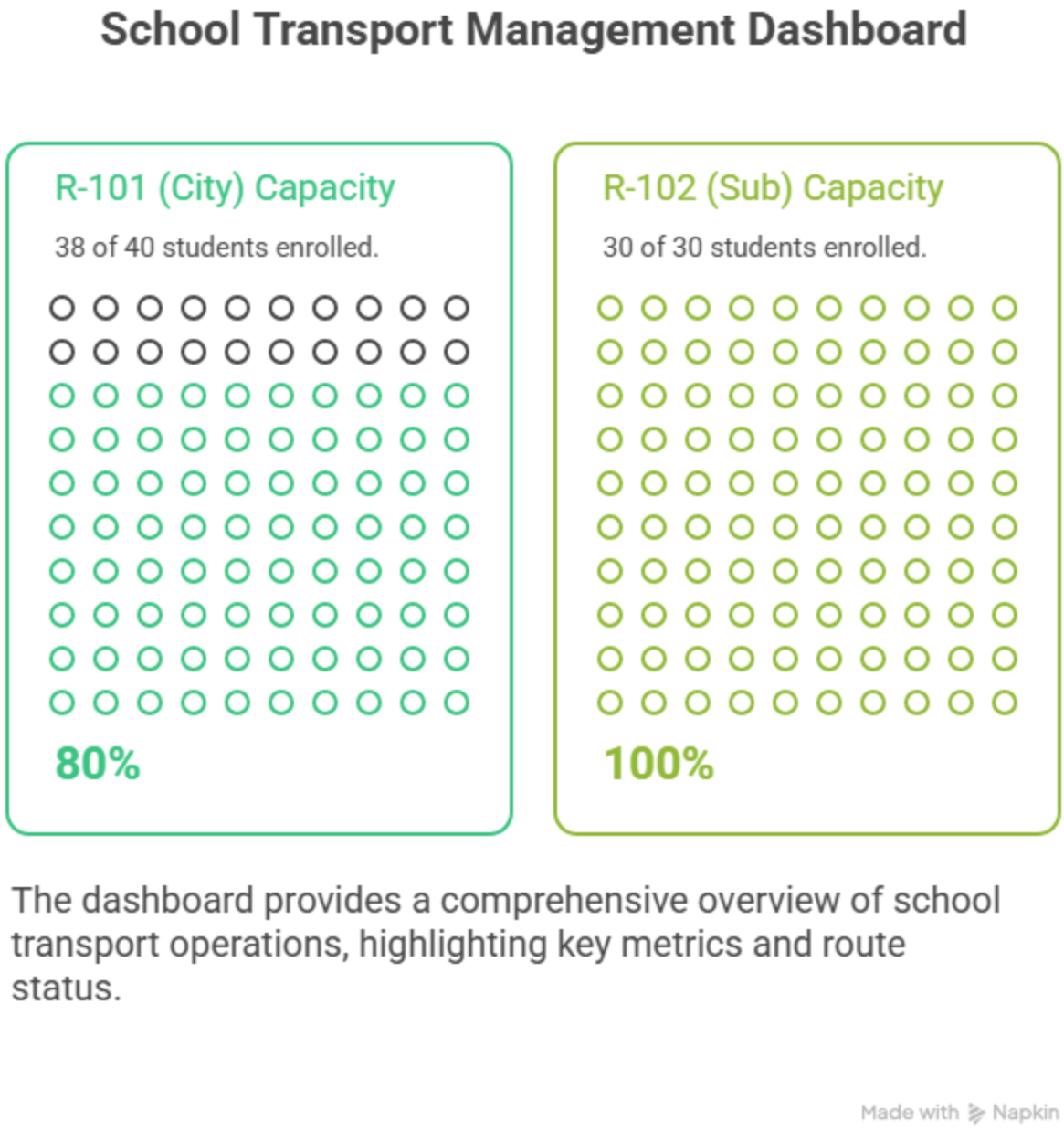


Figure 2. Transport Administration & Route Management Dashboard [cite: 1]

+----------+ | STUDENT TRANSPORT REGISTRATION MODULE | +----------+ | Student ID: [1042] Student Name: John Doe | | Select Route: [R-101 - City Route v] | | Select Stop: [Central Station (Stop #2) v] | | Fee Plan: [Annual Standard (₹15,000) v] | | | | [Register Student] [Cancel] | | Status: Successfully registered. Capacity trigger

passed. | +----------+

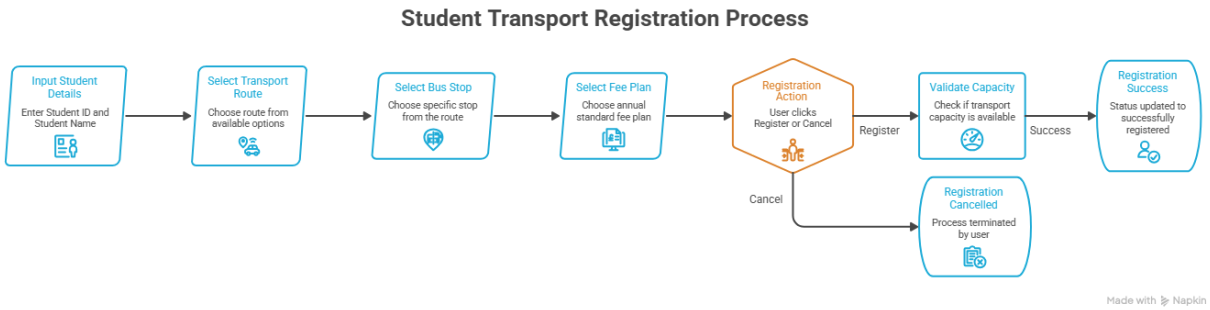


Figure 3. Student Transport Registration & Fee Plan Interface [cite: 1]

+----------+ | DAILY PICKUP ATTENDANCE LOGGING (Driver Portal) | +----------+ | Route: R-101 | Bus: KA-01-F-1234 | Date: 2026-10-07 | |----------+ | Stop Name | Student Name | Status | Action | | Central Station | John Doe | [PRESENT v] | [Update] | | Market Square | Jane Smith | [ABSENT v] | [Update] | +----------+

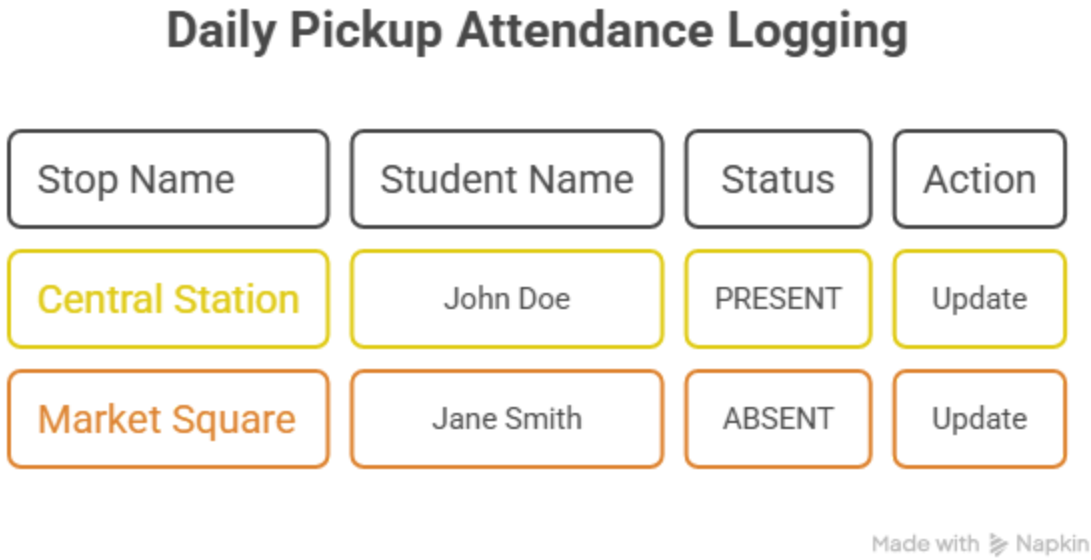


Figure 4. Daily Pickup Attendance & Driver Execution Screen [cite: 1]

+----------+ | CAPACITY UTILIZATION & FINANCIAL SUMMARY REPORT | +----------+ | Route Name |

Capacity | Enrolled | Utilization % | Fee Due| R-101 | 40 | 38 | 95.0% | ₹0 | | R-102 | 30 | 30 | 100.0% | ₹15,000| |----------+ | [Export to CSV] [Print Report] | +----------+

Figure 5. Capacity Utilization & Financial Summary Report Output [cite: 1]

3.8 Testing and Results

Table 15. Test Cases [cite: 1]

| Test ID [cite: 1] | Feature [cite: 1] | Input / Action [cite: 1] | Expected Result [cite: 1] | Actual Result [cite: 1] | Status [cite: 1] |

| :--- | :--- | :--- | :--- | :--- | :--- |

| TC-01 [cite: 1] | Capacity Trigger [cite: 3] | Insert registration exceeding bus capacity [cite: 3] | Rejected by trigger [cite: 3] | Blocked with SQLSTATE '45000' [cite: 3] | Pass [cite: 1] |

| TC-02 [cite: 1] | Driver License [cite: 3] | Assign driver with expired license [cite: 3] | Rejected by validation [cite: 3] | Assignment rejected successfully [cite: 3] | Pass [cite: 1] |

| TC-03 [cite: 1] | Duplicate Pickup [cite: 3] | Insert two attendance logs for one student on same date [cite: 3] | Rejected by UNIQUE constraint [cite: 3] | Duplicate entry prevented [cite: 3] | Pass [cite: 1] |

| TC-04 [cite: 1] | Route-Stop Mapping [cite: 2, 3] | Register student to stop not on selected route [cite: 3] | Rejected by composite FK [cite: 3] | Foreign key violation caught [cite: 3] | Pass [cite: 1] |

Key Results and Observations: The rigorous testing phase confirmed that all 13 normalized relations, declarative constraints, and procedural triggers correctly enforce institutional business rules entirely within the database layer [cite: 2, 3]. The application successfully interfaces with MySQL 8.0, maintaining data integrity and generating real-time operational reports [cite: 2, 3].

---CHAPTER 4: CONCLUSION

This project successfully designed, normalized, and implemented a comprehensive relational database management system for school transport route and student pickup operations [cite: 4]. By structuring the schema up to Third Normal Form (3NF) across 13 tables, data redundancy and update anomalies were successfully eliminated [cite: 2, 3]. Critical business rules—such as bus capacity limits, driver license validity, and stop-route matching—were robustly enforced using database triggers and foreign key constraints [cite: 2, 3]. Furthermore, stored procedures, role-based views, and custom reporting modules delivered an efficient administrative solution [cite: 3].

4.1 Limitations and Future Enhancements

* **Limitation 1:** The current application relies on manual attendance logging by drivers/attendants rather than automated RFID/biometric scanning.

* **Limitation 2:** Real-time GPS coordinates of buses are not dynamically tracked within the relational database [cite: 4].

* **Future Enhancement:** Integration of IoT-based RFID student card readers and real-time GPS tracking hardware to provide live bus tracking for parents [cite: 3].

REFERENCES

[1] A. A. Author, *Database Systems: Concepts and Design*, 3rd ed. City, Country: Publisher, 2025 [cite: 1].

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[3] A. A. Author and B. B. Author, “A database approach for transaction management,” *Journal of Database Research*, vol. 12, no. 2, pp. 101-112, Apr. 2025, doi: 10.xxxx/xxxxx [cite: 1].

[4] Organization Name, *Database Application Security Guidelines*, Tech. Rep. TR-2025-04, City, Country, 2025 [cite: 1].

APPENDICES

APPENDIX A - COMPLETE DATABASE SCRIPT

```sql

-- Complete MySQL 8.0 Script Reference

-- Includes DDL definitions, constraints, triggers, and analytical views for School Transport DBMS.

-- Refer to submitted file: School_Transport_DBMS_Complete_Script.sql

```[cite: 1, 2, 3]

APPENDIX B - APPLICATION SOURCE CODE

```text

Source Code Repository Reference: Local Git Repository / Department Server

```[cite: 1, 3]

APPENDIX C - ADDITIONAL TEST DATA AND OUTPUTS

```text

Additional test execution logs, CSV report exports, and database backup snapshots are stored in the project supplementary archive folder.

```[cite: 1]